TauTimp — Cylindrical Symmetry: Generalized Variable-Coefficient Wave Equation

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Parameter: $R = 3.9228 \times 10^{22}$ m

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1. Master Field Equation in Cylindrical Coordinates

The TauTimp field equation in cylindrical coordinates (r, θ, z) with axial symmetry along z:

$$\frac{R}{r}\,\Delta H(r,\theta,z,t) = \frac{1}{c^2}\frac{\partial^2 H}{\partial t^2}$$

The cylindrical Laplacian with **azimuthal symmetry** ($\partial/\partial\theta = 0$):

$$\Delta = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial}{\partial r}\right) + \frac{\partial^2}{\partial z^2} = \frac{\partial^2}{\partial r^2} + \frac{1}{r}\frac{\partial}{\partial r} + \frac{\partial^2}{\partial z^2}$$

The full master equation becomes:

$$\boxed{\frac{R}{r}\left(\frac{\partial^2 f}{\partial r^2} + \frac{1}{r}\frac{\partial f}{\partial r} + \frac{\partial^2 f}{\partial z^2}\right) = \frac{1}{c^2}\frac{\partial^2 f}{\partial t^2}}$$

2. Separation of Variables

Let:

$$f(r,z,t) = H_3(r)\,H_4(z)\,u(t)$$

Substituting into the master equation and dividing by $H_3\,H_4\,u$:

$$\frac{R}{r}\left[\frac{H_3'' + H_3'/r}{H_3} + \frac{H_4''}{H_4}\right] = \frac{u''}{c^2\,u}$$

The right side depends only on t; the left side depends on r and z. Both equal a separation constant $-\omega^2$:

$$\frac{u''}{c^2\,u} = -\omega^2 \Longrightarrow u'' + \omega^2 c^2\,u = 0$$

and:

$$\frac{R}{r}\left[\frac{H_3'' + H_3'/r}{H_3} + \frac{H_4''}{H_4}\right] = -\omega^2$$

Now separate the radial and axial parts with a second constant k_z^2:

$$\frac{H_4''}{H_4} = -k_z^2$$

$$\frac{R}{r}\cdot\frac{H_3'' + H_3'/r}{H_3} = -\omega^2 + \frac{R\,k_z^2}{r}$$

3. Three Separated ODEs 3.1 Temporal ODE (T)

$$u'' + \Omega^2\,u = 0, \qquad \Omega = \omega c$$

$$\boxed{u(t) = A\cos(\Omega t) + B\sin(\Omega t)}$$

3.2 Axial ODE (Z) — Sinusoidal $H_4(z)$

$$H_4'' + k_z^2\,H_4 = 0$$

$$\boxed{H_4(z) = C_1\cos(k_z z) + C_2\sin(k_z z)}$$

The condition that H_4 be **sinusoidal** (as specified) is satisfied for $k_z^2 > 0$, confirming propagating axial modes.

3.3 Radial ODE (R)

Equation (R) rearranges to:

$$H_3'' + \frac{1}{r}H_3' + \left(\frac{\omega^2 r}{R} - k_z^2\right)H_3 = 0$$

Define the **effective radial wave number**:

$$\kappa^2(r) = \frac{\omega^2 r}{R} - k_z^2$$

This is a **Bessel equation** with position-dependent argument. Introduce $\xi = \xi(r)$ via:

$$\frac{d^2 H_3}{dr^2} + \frac{1}{r}\frac{dH_3}{dr} + \left(\frac{\omega^2 r}{R} - k_z^2\right)H_3 = 0$$

4. Asymptotic Reduction: Radial Airy Structure 4.1 Dominant Balance Analysis

For the regime $\omega^2 r/R \gg k_z^2$ (i.e. $r \gg k_z^2 R/\omega^2$), equation (R-ODE) approaches:

$$H_3'' + \frac{1}{r}H_3' + \frac{\omega^2 r}{R}\,H_3 = 0$$

Substitute $H_3(r) = r^{-1/2}\,w(r)$:

$$w'' + \left(\frac{\omega^2 r}{R} - \frac{1}{4r^2}\right)w = 0$$

The $1/(4r^2)$ term is negligible for large r. Applying the Langer transformation $\xi = -(\omega^2/R)^{1/3}\, r$:

$$\frac{d^2w}{d\xi^2} = \xi\, w$$

This is the **Airy equation**, with solution $w = C_1\,\mathrm{Ai}(\xi) + C_2\,\mathrm{Bi}(\xi)$.
Regularity at large r (boundedness) requires $C_2 = 0$.

4.2 General Radial Solution (with k_z correction)

The full equation (R-ODE) with both terms is equivalent — via $s = 2\omega r^{1/2}/(R^{1/4}\cdot r^{1/4})$ — to a **modified Bessel equation**. For the combined case write:

$$H_3(r) = r^{-1/2}[C_1\,\mathrm{Ai}(-\mu r + \nu) + C_2\,\mathrm{Bi}(-\mu r + \nu)]$$

where:

$$\mu = \left(\frac{\omega^2}{R}\right)^{1/3}, \qquad \nu = \left(\frac{\omega^2}{R}\right)^{-1/3} \frac{k_z^2}{1}\cdot\frac{1}{\mu} = \frac{k_z^2}{\mu^2} = \frac{k_z^2 R^{2/3}}{\omega^{4/3}}$$

The effective Airy variable is:

$$\zeta(r) = \mu r - \nu = \left(\frac{\omega^2}{R}\right)^{1/3} r - \frac{k_z^2\, R^{2/3}}{\omega^{4/3}}$$

Physical solution ($C_2 = 0$):

$$\boxed{H_3(r) = \frac{C}{\sqrt{r}}\,\mathrm{Ai}(-\zeta(r))}$$

5. Boundary Condition $H_3(R) = 0$ and Eigenvalue Quantization

The condition $H_3(R) = 0$ requires:

$$\mathrm{Ai}(-\zeta(R)) = 0$$

Denoting the n-th positive zero of $\mathrm{Ai}(-x)$ as a_n (with $a_1 \approx 2.3381$, $a_2 \approx 4.0879$, ...):

$$\zeta(R) = a_n \Longrightarrow \left(\frac{\omega_n^2}{R}\right)^{1/3} R - \frac{k_z^2\, R^{2/3}}{\omega_n^{4/3}} = a_n$$

Define $\alpha_n = \omega_n^{2/3}/R^{1/3}$. Then:

$$\alpha_n\, R - \frac{k_z^2}{\alpha_n^2} = a_n$$

This gives the **implicit eigenvalue relation**:

$$\alpha_n^3\, R - k_z^2 = a_n\, \alpha_n^2$$

$$\boxed{\omega_n^2 = R\left(\frac{a_n + \sqrt{a_n^2 + 4k_z^2 R}}{2R}\right)^3 \cdot \frac{1}{1}}$$

For the $k_z = 0$ **case** (axially uniform):

$$\omega_n|_{k_z=0} = \sqrt{R}\left(\frac{a_n}{R}\right)^{3/2} = \frac{a_n^{3/2}}{R}$$

$$\boxed{k_n|_{k_z=0} = \frac{\omega_n}{c} = \frac{a_n^{3/2}}{R\,c}}$$

For general k_z the eigenfrequency is:

$$\omega_n(k_z) = \left[R\left(\frac{a_n}{R} + k_z^{2/3}\cdot\frac{R^{1/3}}{a_n^{1/3}}\right)^{3/2}\right]^{1/2}$$

(asymptotic form; exact value from cubic above).

6. Complete Field Solution $f(r,z,t)$ Mode structure

Each mode is labeled (n,m) with $n \geq 1$ (radial Airy index) and $m \in \mathbb{Z}$ (axial index, $k_z = m\pi/L$ for axial domain $[0,L]$):

$$f_{nm}(r,z,t) = \frac{1}{\sqrt{r}}\,\mathrm{Ai}(-\zeta_n(r))\cdot\left[C_1^{(m)}\cos(k_z^{(m)}z) + C_2^{(m)}\sin(k_z^{(m)}z)\right]\cdot[A_{nm}\cos(\Omega_{nm}t) + B_{nm}\sin(\Omega_{nm}t)]$$

General solution (superposition)

$$\boxed{f(r,z,t) = \sum_{n=1}^{\infty}\sum_{m=-\infty}^{\infty}\frac{1}{\sqrt{r}}\,\mathrm{Ai}(-\zeta_n(r))\left[D_{nm}^{(c)}\cos(k_m z) + D_{nm}^{(s)}\sin(k_m z)\right][A_{nm}\cos(\Omega_{nm}t) + B_{nm}\sin(\Omega_{nm}t)]}$$

Coefficients fixed by initial/boundary conditions via orthogonality of Ai modes and Fourier series in z.

7. Full Angular Field $H(r,\theta,z,t)$

Since f is already azimuthally symmetric, the full field with higher azimuthal modes $\theta^m = e^{im\theta}$ is:

$$H(r,\theta,z,t)=\sum_{m_\theta=0}^{\infty}e^{im_\theta\theta}\sum_{n=1}^{\infty}\sum_{m_z}\frac{D_{n,m_\theta,m_z}}{\sqrt{r}}\,\mathrm{Ai}\left(-\zeta_n^{(m_\theta)}(r)\right)e^{ik_{m_z}z}[A_{nm}\cos(\Omega_{nm}t)+B_{nm}\sin(\Omega_{nm}t)]$$

For the **axially symmetric,** $m_\theta=0$ **sector** this reduces to $f(r,z,t)$ above.

8. Photon Probability Density — Squared-Amplitude Formalism 8.1 Definitions

The photon probability density in cylindrical symmetry:

$$P(r,z,t)=|\psi(r,z,t)|^2$$

satisfying the **radial wave equation** (d'Alembert in r at fixed z,t):

$$\frac{\partial^2 P}{\partial t^2}-c^2\frac{\partial^2 P}{\partial r^2}=0$$

This is the **1D wave equation** in r with speed c (or V).

8.2 General Solution

$$P(r,z,t)=F(r-Vt,\,z)+G(r+Vt,\,z)$$

The z-dependence enters as a parameter (since the equation acts only in r,t).

8.3 Boundary Condition $P(R,z,0)=0$

At $t=0$: $F(R,z)+G(R,z)=0\Rightarrow G(R,z)=-F(R,z)$.

The reflection principle about $r=R$ gives $G(s,z)=-F(2R-s,z)$, yielding **standing-wave eigenmodes**:

$$P_n(r,z,t)=\mathcal{A}_n(z)\sin\left(\frac{n\pi r}{R}\right)\cos\left(\frac{n\pi Vt}{R}\right),\qquad \omega_n^{(P)}=\frac{n\pi V}{R}$$

The axial profile $\mathcal{A}_n(z)$ is inherited from the initial condition on $P(r,z,0)$.

8.4 Velocity Variants — $t=0$ snapshot

At $t=0$, $P(r,z,0)=\mathcal{A}(z)\sin(n\pi r/R)$ for each V. The spatial profile is **identical** for all velocities — only the temporal frequency $\omega_n=n\pi V/R$ changes:

Variant	V [m/s]	$\omega_1^{(P)}$ [rad/s]	$T_1^{(P)}$ [s]
Light	$c=2.998\times10^{8}$	2.404×10^{-14}	2.614×10^{14}
Onduscular	10^9c	2.404×10^{-5}	2.614×10^{5}

8.5 Full Probability Density (cylindrical, z-dependent)

Including the sinusoidal z-structure from $H_4(z)$:

$$P(r,z,t)=\sum_n\sum_m\sin\left(\frac{n\pi r}{R}\right)[C_{nm}\cos(k_mz)+S_{nm}\sin(k_mz)]\cos\left(\frac{n\pi Vt}{R}\right)$$

8.6 Physical flux observable

The 3D probability current density:

$$\vec{J}(r,z,t)=-\frac{c^2}{2}\nabla P=-\frac{c^2}{2}\left(\hat{r}\frac{\partial P}{\partial r}+\hat{z}\frac{\partial P}{\partial z}\right)$$

The **radial flux through a cylinder of radius** r**, length** L_z:

$$\Phi(r,t)=2\pi r\int_0^{L_z}J_r\,(r,z,t)\,dz$$

9. Generalized Parameter Table

$$R=3.9228\times10^{22}\text{ m},\qquad c=2.998\times10^{8}\text{ m/s}$$

Airy eigenfrequencies $\omega_n(k_z=0)=a_n^{3/2}/R\cdot c$ (photon, $V=c$)

n	a_n	$k_n^{(r)}$ [m^{-1}]	ω_n [rad/s]
1	2.3381	3.578×10^{-23}	1.073×10^{-14}
2	4.0879	8.297×10^{-23}	2.488×10^{-14}
3	5.5206	1.308×10^{-22}	3.923×10^{-14}
4	6.7867	1.780×10^{-22}	5.338×10^{-14}

d'Alembert frequencies $\omega_n=n\pi V/R$

n	$V=c$	$V=10^9c$
1	2.404×10^{-14}	2.404×10^{-5}
2	4.808×10^{-14}	4.808×10^{-5}
3	7.213×10^{-14}	7.213×10^{-5}

10. Asymptotic WKB Analysis

For $\zeta(r)\gg1$:

$$\mathrm{Ai}(-\zeta)\approx\frac{1}{\sqrt{\pi}\,\zeta^{1/4}}\sin\left(\frac{2}{3}\zeta^{3/2}+\frac{\pi}{4}\right)$$

The radial eigenfunction has **local wave number**:

$$k_{\rm eff}(r) = \frac{d}{dr}\left(\frac{2}{3}\zeta^{3/2}\right) = \sqrt{\zeta(r)}\cdot\mu = \sqrt{\frac{\omega^2 r}{R}-k_z^2}$$
The **phase speed** varies as:
$$V_{\rm ph}(r) = \frac{\Omega}{k_{\rm eff}(r)} = \frac{\omega c}{\sqrt{\omega^2 r/R - k_z^2}} = \frac{c}{\sqrt{r/R - k_z^2 R/(\omega^2 R^2)}}$$
At $r = R$: $V_{\rm ph}(R) = c/\sqrt{1 - k_z^2/k_n^2}$ — shows sub-luminal phase velocity for $k_z \neq 0$.

11. Normalization and Orthogonality

The radial orthogonality (weight function r):

$$\int_0^R H_3^{(n)}(r)\, H_3^{(m)}(r)\, r\, dr = \mathcal{N}_n\, \delta_{nm}$$

$$\mathcal{N}_n = \int_0^R \frac{1}{r}[\mathrm{Ai}(-\zeta_n(r))]^2 r\, dr = \int_0^R [\mathrm{Ai}(-\zeta_n(r))]^2\, dr$$

Changing variable $\rho = \zeta_n(r) = \mu_n r$, $dr = d\rho/\mu_n$:

$$\mathcal{N}_n = \frac{1}{\mu_n}\int_0^{a_n} [\mathrm{Ai}(-\rho)]^2\, d\rho = \frac{R^{1/3}}{\omega_n^{2/3}}\int_0^{a_n} [\mathrm{Ai}(-\rho)]^2\, d\rho$$

The integral $I_n = \int_0^{a_n}[\mathrm{Ai}(-\rho)]^2\, d\rho$ is computed numerically: $I_1 \approx 0.2456$, $I_2 \approx 0.2148$, $I_3 \approx 0.2041$.

12. Summary of Complete Solution

Component	Expression
Radial Airy eigenfunction	$H_3^{(n)}(r) = r^{-1/2}\mathrm{Ai}(-\zeta_n(r))$
Airy variable	$\zeta_n(r) = (\omega_n^2/R)^{1/3} r - k_z^2 R^{2/3}/\omega_n^{4/3}$
BC quantization	$\zeta_n(R) = a_n$ (Airy zeros)
Axial factor	$H_4(z) = C_1\cos(k_z z) + C_2\sin(k_z z)$
Temporal factor	$u(t) = A\cos(\Omega t) + B\sin(\Omega t)$, $\Omega = \omega_n c$
Full Airy field	$f(r,z,t) = \sum H_3^{(n)} H_4 u$
d'Alembert mode	$P_n = \sin(n\pi r/R)\cos(n\pi V t/R)$
Angular extension	$H = \sum f_{nm}(r,z,t)\, e^{im\theta}$

Appendix: Physical Scales

$$R = 3.9228 \times 10^{22}\ \mathrm{m} \approx 1.272\ \mathrm{kpc}$$
$$\mu_1 = (\omega_1^2/R)^{1/3}, \quad a_1 = 2.3381$$
$$k_1^{(r)} = a_1^{3/2}/R \approx 3.578 \times 10^{-23}\ \mathrm{m}^{-1}$$
$$\lambda_1 = 2\pi/k_1^{(r)} \approx 1.756 \times 10^{23}\ \mathrm{m} \approx 5.69\ \mathrm{kpc}$$

See companion file TauTimp_CylindricalSymmetry.mpl for all numerical computations and 13 structured plots.

Plot